

Classification_Decision_Tree

November 22, 2025

```
[1]: # In COLAB, we will now use sklearn.metrics for plotting ROC and
      ↳ Precision-Recall curves instead of scikit-plot for compatibility.
      # No specific installations are needed for scikit-learn's plotting functions as
      ↳ it's typically pre-installed.
```

```
[2]: import numpy as np
      import pandas as pd
      import seaborn as sns
      import matplotlib.pyplot as plt
```

Titanic Dataset

```
[3]: df = pd.read_csv("data/titanic.csv")
      df.head()
```

```
[3]: PassengerId  Survived  Pclass  \
0             1         0         3
1             2         1         1
2             3         1         3
3             4         1         1
4             5         0         3
```

```

                                Name    Sex  Age  SibSp  \
0                Braund, Mr. Owen Harris  male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1
2                Heikkinen, Miss. Laina  female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0    1
4                Allen, Mr. William Henry   male  35.0    0
```

```

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0 STON/O2. 3101282   7.9250   NaN      S
3      0    113803  53.1000  C123      S
4      0   373450   8.0500   NaN      S
```

Data Cleaning

```
[4]: df.isnull().sum()
```

```
[4]: PassengerId      0
      Survived        0
      Pclass         0
      Name           0
      Sex            0
      Age           177
      SibSp          0
      Parch          0
      Ticket         0
      Fare           0
      Cabin         687
      Embarked       2
      dtype: int64
```

```
[5]: df['Age'] = df.groupby(['Sex', 'Pclass'])['Age'].transform(lambda x: x.fillna(x.
      ↪median()))
```

```
[6]: df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

Removing useless variables

```
[7]: column2drop = ['PassengerId', 'Name', 'Cabin']
      df.drop(column2drop, axis=1, inplace=True)
```

```
[8]: df.head()
```

```
[8]:
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	\
0	0	3	male	22.0	1	0	A/5 21171	7.2500	
1	1	1	female	38.0	1	0	PC 17599	71.2833	
2	1	3	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	1	1	female	35.0	1	0	113803	53.1000	
4	0	3	male	35.0	0	0	373450	8.0500	

```
      Embarked
0          S
1          C
2          S
3          S
4          S
```

0.0.1 Feature Engineering

```
[9]: # number of people a passanger is travelling with
      df['FamilySize'] = df['SibSp'] + df['Parch'] + 1
```

```
[10]: # if he/she is travelling alone or not
df['IsAlone'] = 1 # initialize to yes/1 is alone
df.loc[df.FamilySize > 1, 'IsAlone'] = 0 # update to no/0 if family size is
↳ greater than 1
```

Features SibSp and Parch are now redundant with respect to FamilySize.

```
[11]: column2drop = ['SibSp', 'Parch']
df.drop(column2drop, axis=1, inplace=True)
```

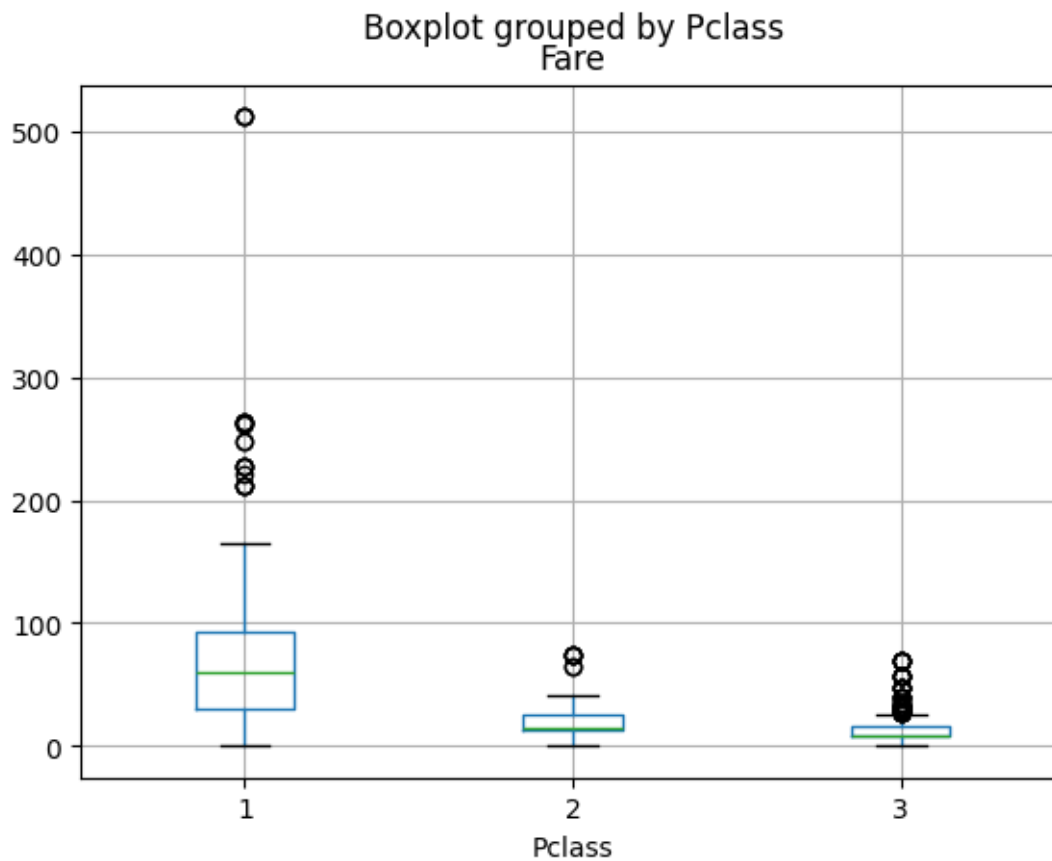
```
[12]: df.head()
```

```
[12]:
```

	Survived	Pclass	Sex	Age	Ticket	Fare	Embarked	\
0	0	3	male	22.0	A/5 21171	7.2500	S	
1	1	1	female	38.0	PC 17599	71.2833	C	
2	1	3	female	26.0	STON/O2. 3101282	7.9250	S	
3	1	1	female	35.0	113803	53.1000	S	
4	0	3	male	35.0	373450	8.0500	S	

	FamilySize	IsAlone
0	2	0
1	2	0
2	1	1
3	2	0
4	1	1

```
[13]: # examining Fare feature
df.boxplot(['Fare'], by='Pclass', showfliers=True)
plt.show()
```



```
[14]: dupl_df = df[df.duplicated('Ticket', keep=False)]
      dupl_df
```

```
[14]:
```

	Survived	Pclass	Sex	Age	Ticket	Fare	Embarked	FamilySize	\
3	1	1	female	35.0	113803	53.1000	S	2	
7	0	3	male	2.0	349909	21.0750	S	5	
8	1	3	female	27.0	347742	11.1333	S	3	
9	1	2	female	14.0	237736	30.0708	C	2	
10	1	3	female	4.0	PP 9549	16.7000	S	3	
..	...	...	...	...	...	...	...		
876	0	3	male	20.0	7534	9.8458	S	1	
879	1	1	female	56.0	11767	83.1583	C	2	
880	1	2	female	25.0	230433	26.0000	S	2	
885	0	3	female	39.0	382652	29.1250	Q	6	
888	0	3	female	21.5	W./C. 6607	23.4500	S	4	

	IsAlone
3	0
7	0

```

8          0
9          0
10         0
..         ...
876        1
879        0
880        0
885        0
888        0

```

[344 rows x 9 columns]

```
[15]: std = dupl_df[['Fare', 'Ticket']].groupby(['Ticket']).std()
std
```

```
[15]:
Ticket
110152    0.0
110413    0.0
110465    0.0
111361    0.0
113505    0.0
...
SC/Paris 2123    0.0
STON/O2. 3101279    0.0
W./C. 6607    0.0
W./C. 6608    0.0
WE/P 5735    0.0

```

[134 rows x 1 columns]

```
[16]: std[std['Fare'] > 0]
```

```
[16]:
Ticket
7534    0.444841

```

```
[17]: df[df['Ticket'] == '7534']
```

```
[17]:
Survived  Pclass  Sex  Age  Ticket  Fare  Embarked  FamilySize  IsAlone
138      0      3  male  16.0  7534  9.2167      S      1      1
876      0      3  male  20.0  7534  9.8458      S      1      1

```

```
[18]: df[df['Ticket'] == '110152']
```

```
[18]:
Survived  Pclass  Sex  Age  Ticket  Fare  Embarked  FamilySize  \
257      1      1  female  30.0  110152  86.5      S      1

```

504	1	1	female	16.0	110152	86.5	S	1
759	1	1	female	33.0	110152	86.5	S	1

IsAlone	
257	1
504	1
759	1

It looks like Fare is cumulative with respect to various Tickets with the same identifier. Consequently we proceed in correcting the Fare of each passenger by dividing it with the number of Tickets with the same identifier.

```
[19]: ticket_map = dict(df['Ticket'].value_counts()) # n. of occurrences by ticket
df['NumTickets'] = df['Ticket'].map(ticket_map) # creates new column based on
↳ the value counts map
```

```
[20]: df['Fare'] = df['Fare']/df['NumTickets']
```

```
[21]: df[df['Ticket'] == '110152']
```

```
[21]:
```

	Survived	Pclass	Sex	Age	Ticket	Fare	Embarked	FamilySize	\
257	1	1	female	30.0	110152	28.833333	S	1	
504	1	1	female	16.0	110152	28.833333	S	1	
759	1	1	female	33.0	110152	28.833333	S	1	

	IsAlone	NumTickets
257	1	3
504	1	3
759	1	3

The features NumTickets and Ticket helped in correcting Fare and we do not need them any more.

```
[22]: column2drop = ['NumTickets', 'Ticket']
df.drop(column2drop, axis=1, inplace=True)
```

```
[23]: df.head()
```

```
[23]:
```

	Survived	Pclass	Sex	Age	Fare	Embarked	FamilySize	IsAlone
0	0	3	male	22.0	7.2500	S	2	0
1	1	1	female	38.0	71.2833	C	2	0
2	1	3	female	26.0	7.9250	S	1	1
3	1	1	female	35.0	26.5500	S	2	0
4	0	3	male	35.0	8.0500	S	1	1

```
[24]: from sklearn.preprocessing import LabelEncoder
```

```
[25]: # label encoding Sex feature
le = LabelEncoder()
```

```
df['Sex'] = le.fit_transform(df['Sex'])
```

```
[26]: # one-hot encoding of categorical data
categorical_cols = ["Embarked"]
df = pd.get_dummies(df, columns=categorical_cols, dtype=int)
```

```
[27]: df.head()
```

```
[27]:
```

	Survived	Pclass	Sex	Age	Fare	FamilySize	IsAlone	Embarked_C	\
0	0	3	1	22.0	7.2500	2	0	0	
1	1	1	0	38.0	71.2833	2	0	1	
2	1	3	0	26.0	7.9250	1	1	0	
3	1	1	0	35.0	26.5500	2	0	0	
4	0	3	1	35.0	8.0500	1	1	0	

	Embarked_Q	Embarked_S
0	0	1
1	0	0
2	0	1
3	0	1
4	0	1

0.0.2 Partitioning

```
[28]: from sklearn.model_selection import train_test_split
```

```
[29]: attributes = [col for col in df.columns if col != 'Survived']
X = df[attributes].values
y = df['Survived']
```

```
[30]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
↳ random_state=1, stratify=y)
```

0.0.3 Decision Tree

```
[31]: from sklearn.tree import DecisionTreeClassifier, plot_tree
```

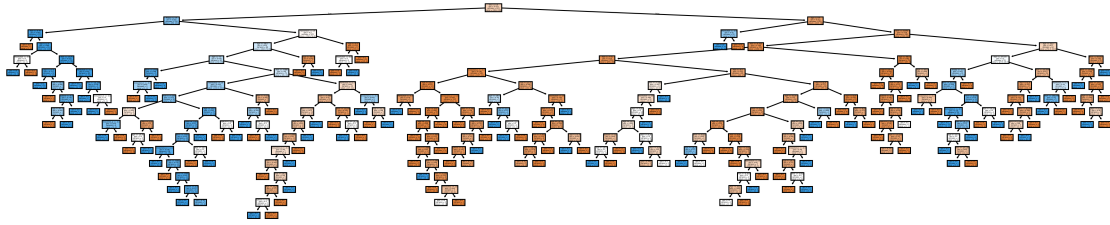
```
[32]: dt = DecisionTreeClassifier()
```

```
[33]: %%time
dt.fit(X_train, y_train)
```

```
CPU times: user 6.3 ms, sys: 0 ns, total: 6.3 ms
Wall time: 10.1 ms
```

```
[33]: DecisionTreeClassifier()
```

```
[34]: plt.figure(figsize=(20, 4), dpi=300)
      plot_tree(dt, feature_names=attributes, filled=True)
      plt.show()
```



```
[35]: from sklearn.metrics import (
      accuracy_score,
      f1_score,
      classification_report,
      confusion_matrix,
      roc_auc_score,
      )
```

```
[36]: y_train_pred = dt.predict(X_train)
      y_test_pred = dt.predict(X_test)
```

```
[37]: print('Train Accuracy %s' % accuracy_score(y_train, y_train_pred))
      print('Train F1-score %s' % f1_score(y_train, y_train_pred, average=None))
      print()

      print('Test Accuracy %s' % accuracy_score(y_test, y_test_pred))
      print('Test F1-score %s' % f1_score(y_test, y_test_pred, average=None))
```

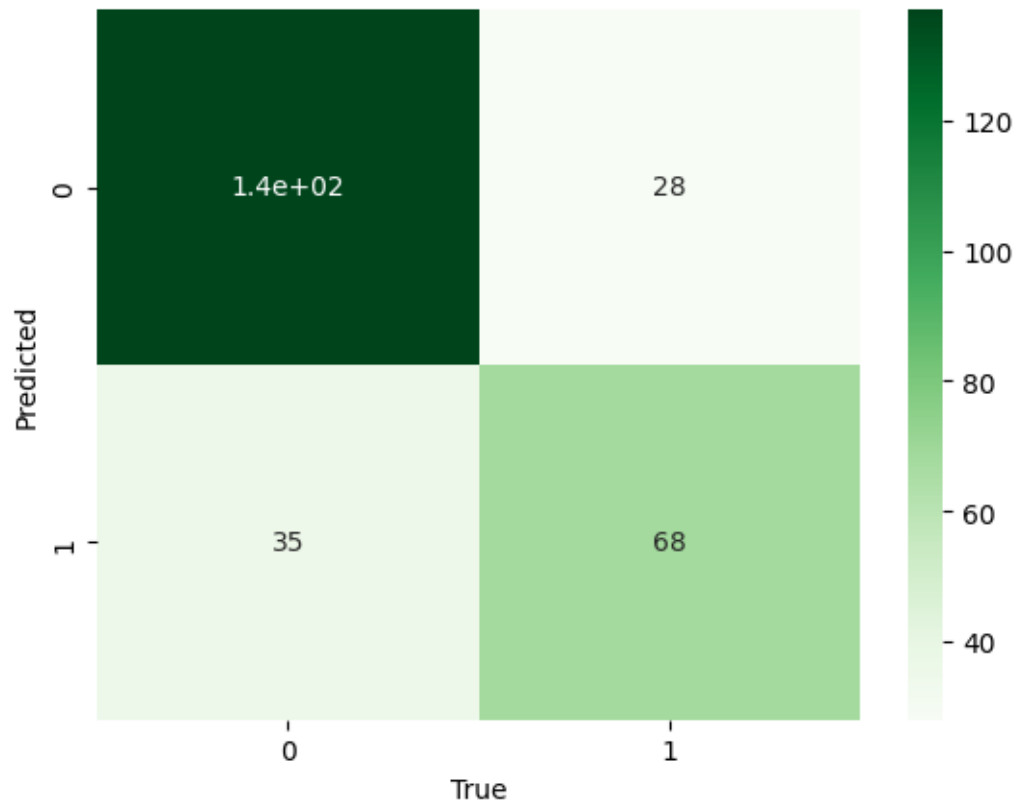
Train Accuracy 0.9839486356340289
 Train F1-score [0.9871134 0.9787234]

Test Accuracy 0.7649253731343284
 Test F1-score [0.81305638 0.68341709]

```
[38]: print(classification_report(y_test, y_test_pred))
```

	precision	recall	f1-score	support
0	0.80	0.83	0.81	165
1	0.71	0.66	0.68	103
accuracy			0.76	268
macro avg	0.75	0.75	0.75	268
weighted avg	0.76	0.76	0.76	268


```
[39]: cf = confusion_matrix(y_test, y_test_pred)
sns.heatmap(cf, annot=True, cmap="Greens")
plt.xlabel("True")
plt.ylabel("Predicted")
plt.show()
```



Feature Importance

```
[40]: zipped = zip(attributes, dt.feature_importances_)
zipped = sorted(zipped, key=lambda x: x[1], reverse=True)
for col, imp in zipped:
    print(col, imp)
```

```
Sex 0.2925058533225051
Fare 0.23954770251754395
Age 0.21692664424863126
FamilySize 0.10357080023756302
Pclass 0.10125791393654211
Embarked_C 0.015174506631692064
Embarked_S 0.01395165145656919
```

```
Embarked_Q 0.009541782631775114
IsAlone 0.007523145017178179
```

Let us discuss overfitting

```
[41]: nodes = []
      train_err = []
      test_err = []

      for max_depth in list(np.arange(1, 20)):
          dt = DecisionTreeClassifier(max_depth=max_depth)
          dt.fit(X_train, y_train)
          y_train_pred = dt.predict(X_train)
          y_test_pred = dt.predict(X_test)

          train_err.append(1 - accuracy_score(y_train, y_train_pred))
          test_err.append(1 - accuracy_score(y_test, y_test_pred))
          nodes.append(dt.tree_.node_count)
```

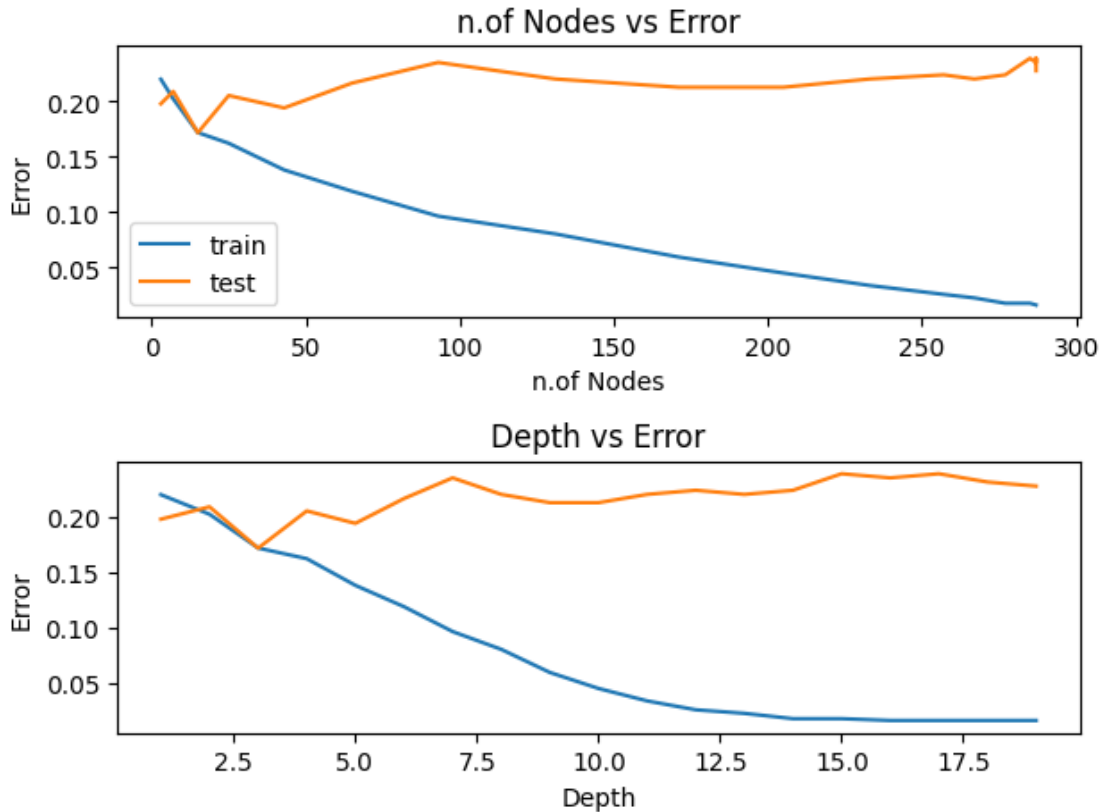
```
[42]: fig, ax = plt.subplots(2,1)
      ax[0].plot(nodes, train_err, label='train')
      ax[0].plot(nodes, test_err, label='test')
      ax[0].legend()

      ax[0].set_xlabel("n.of Nodes")
      ax[0].set_ylabel("Error")
      ax[0].set_title("n.of Nodes vs Error")

      ax[1].plot(list(np.arange(1, 20)), train_err, label='train')
      ax[1].plot(list(np.arange(1, 20)), test_err, label='test')

      ax[1].set_xlabel("Depth")
      ax[1].set_ylabel("Error")
      ax[1].set_title("Depth vs Error")

      plt.tight_layout()
      plt.show()
```



[42] :

Decision Tree hyperparameters `max_depth` : int, default=None The maximum depth of the tree. If None, then nodes are expanded until all leaves are pure or until all leaves contain less than `min_samples_split` samples.

`min_samples_split` : int or float, default=2 The minimum number of samples required to split an internal node:

- If int, then consider ``min_samples_split`` as the minimum number.
- If float, then ``min_samples_split`` is a fraction and ``ceil(min_samples_split * n_samples)`` are the minimum number of samples for each split.

`min_samples_leaf` : int or float, default=1 The minimum number of samples required to be at a leaf node. A split point at any depth will only be considered if it leaves at least `min_samples_leaf` training samples in each of the left and right branches. This may have the effect of smoothing the model, especially in regression.

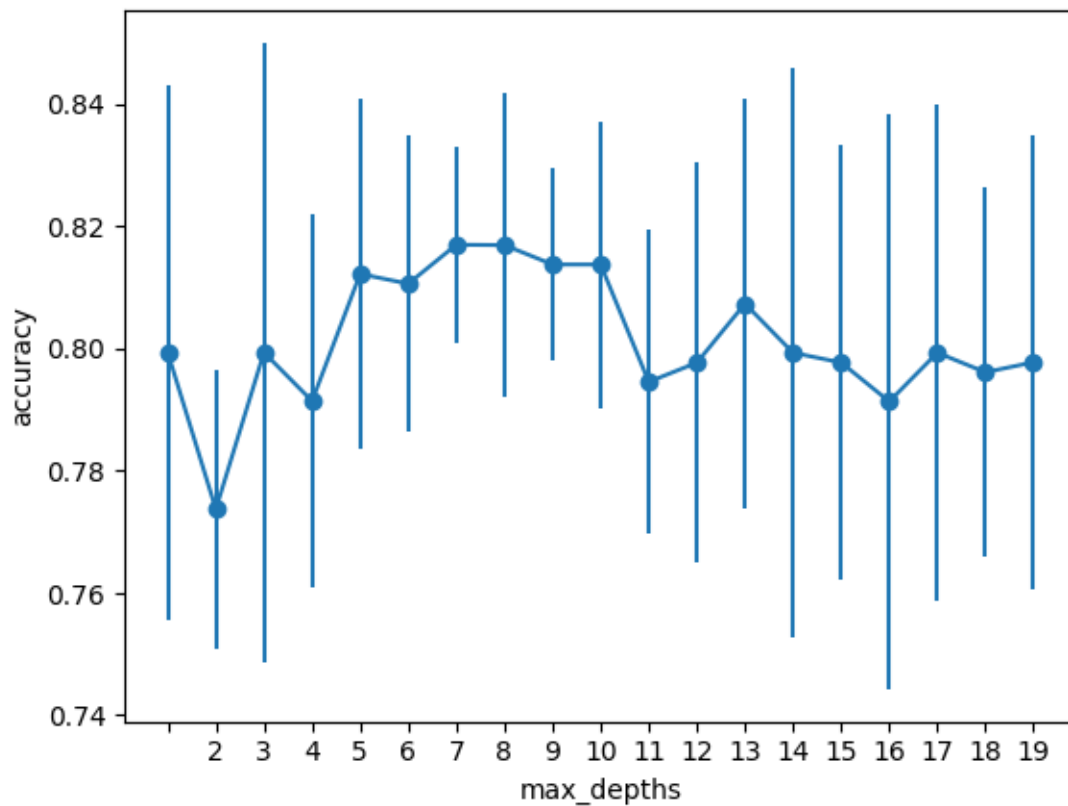
- If int, then consider ``min_samples_leaf`` as the minimum number.
- If float, then ``min_samples_leaf`` is a fraction and ``ceil(min_samples_leaf * n_samples)`` are the minimum number of samples for each node.

```
[43]: from sklearn.model_selection import cross_val_score
```

```
[44]: max_depths = [None] + list(np.arange(2, 20))
avg_scores = list()
std_scores = list()

for max_depth in max_depths:
    dt = DecisionTreeClassifier(max_depth=max_depth)
    scores = cross_val_score(dt, X_train, y_train, cv=5)
    avg_scores.append(np.mean(scores))
    std_scores.append(np.std(scores))

plt.errorbar(range(len(max_depths)), y=avg_scores, yerr=std_scores, marker='o')
plt.xticks(range(len(max_depths)), max_depths)
plt.xlabel("max_depths")
plt.ylabel("accuracy")
plt.show()
```



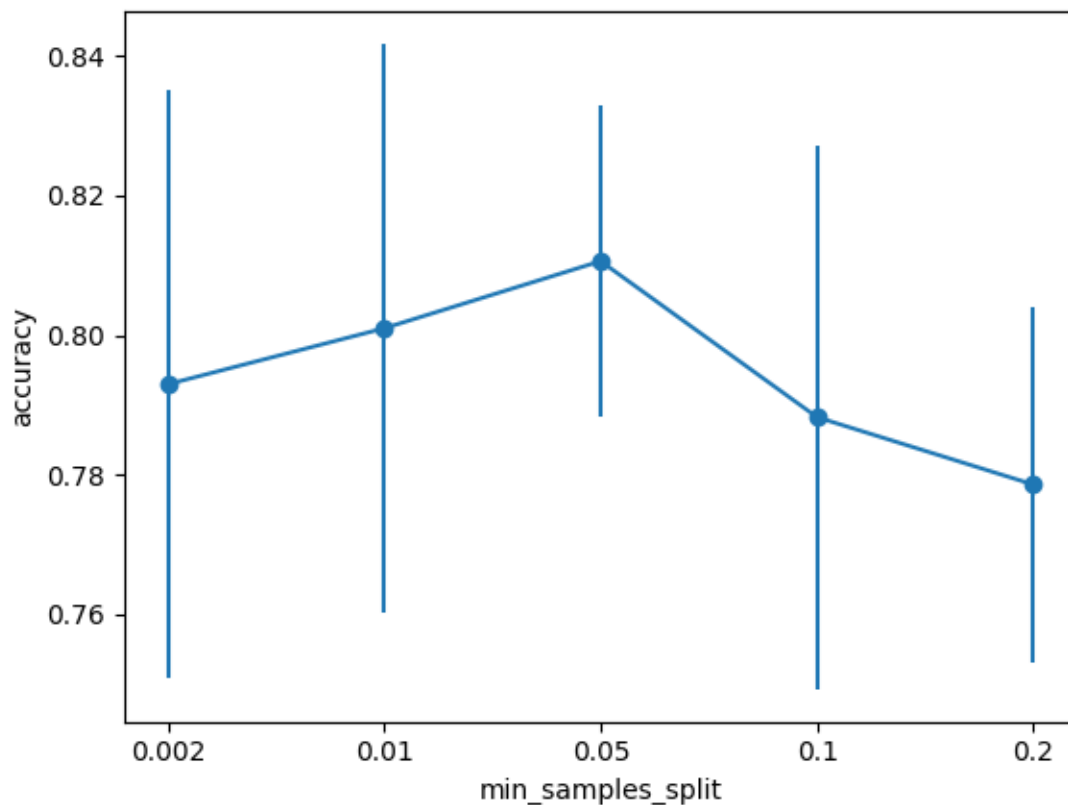
```
[45]: min_samples_splits = [0.002, 0.01, 0.05, 0.1, 0.2]
avg_scores = list()
std_scores = list()
```

```

for min_samples_split in min_samples_splits:
    dt = DecisionTreeClassifier(min_samples_split=min_samples_split)
    scores = cross_val_score(dt, X_train, y_train, cv=5)
    avg_scores.append(np.mean(scores))
    std_scores.append(np.std(scores))

plt.errorbar(range(len(min_samples_splits)), y=avg_scores, yerr=std_scores,
            marker='o')
plt.xticks(range(len(min_samples_splits)), min_samples_splits)
plt.xlabel("min_samples_split")
plt.ylabel("accuracy")
plt.show()

```



```

[46]: min_samples_leafs = [1, 5, 10, 20, 30, 50, 100]
avg_scores = list()
std_scores = list()

for min_samples_leaf in min_samples_leafs:
    dt = DecisionTreeClassifier(min_samples_leaf=min_samples_leaf)
    scores = cross_val_score(dt, X_train, y_train, cv=5)

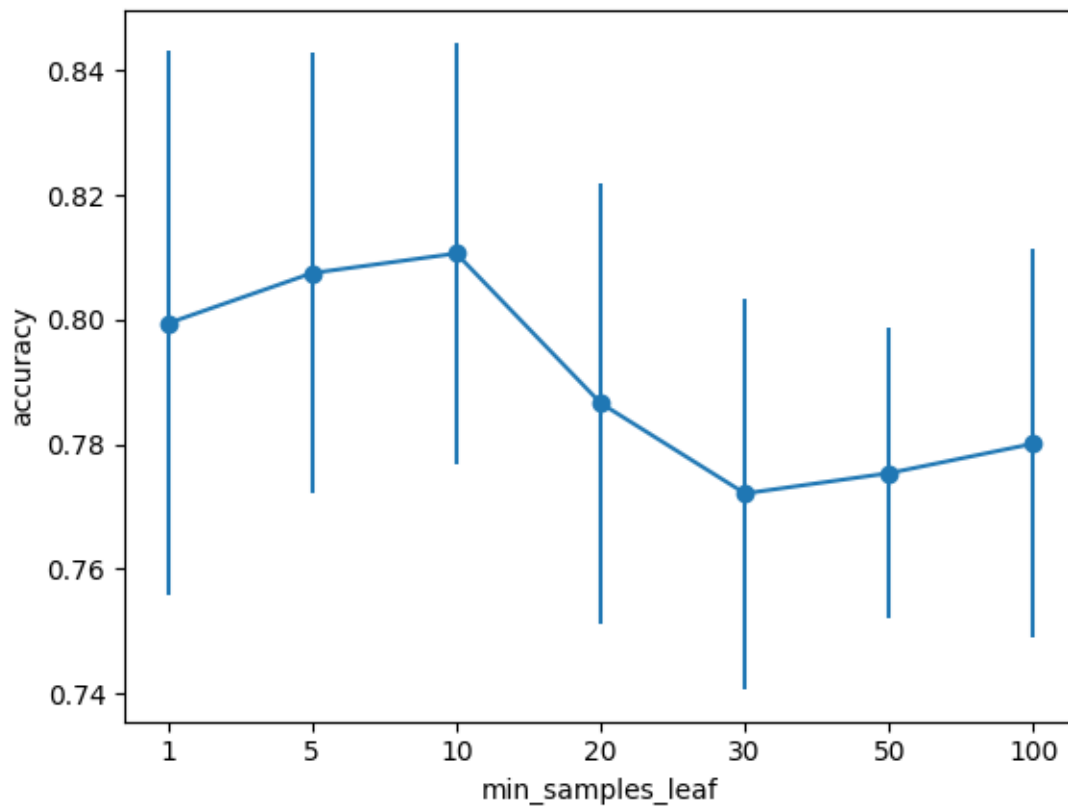
```

```

avg_scores.append(np.mean(scores))
std_scores.append(np.std(scores))

plt.errorbar(range(len(min_samples_leafs)), y=avg_scores, yerr=std_scores,
             marker='o')
plt.xticks(range(len(min_samples_leafs)), min_samples_leafs)
plt.xlabel("min_samples_leaf")
plt.ylabel("accuracy")
plt.show()

```



0.0.4 Randomized Search

```

[47]: from sklearn.model_selection import RandomizedSearchCV
      from sklearn.model_selection import RepeatedStratifiedKFold

```

```

[48]: %%time
      param_list = {
          'max_depth': [None] + list(np.arange(2, 20)),
          'min_samples_split': [2, 5, 10, 20, 30, 50, 100],
          'min_samples_leaf': [1, 5, 10, 20, 30, 50, 100],
          'criterion': ['gini', 'entropy']
      }

```

```

}

clf = DecisionTreeClassifier(criterion='gini', max_depth=None,
    ↪min_samples_split=2, min_samples_leaf=1)

random_search = RandomizedSearchCV(
    DecisionTreeClassifier(),
    param_distributions=param_list,
    cv=RepeatedStratifiedKFold(random_state=0),
    n_jobs=1, # Changed from -1 to 1 to avoid multiprocessing issues
    refit=True,
    n_iter=200,
    #verbose=2
)

random_search.fit(X_train, y_train)
clf = random_search.best_estimator_

```

CPU times: user 35.7 s, sys: 63.1 ms, total: 35.7 s
 Wall time: 36.1 s

```
[49]: print(random_search.best_params_, random_search.best_score_)
```

```
{'min_samples_split': 10, 'min_samples_leaf': 5, 'max_depth': 6, 'criterion':
'entropy'} 0.8200645161290322
```

```
[50]: y_test_pred = clf.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_test_pred))
```

Accuracy: 0.7985074626865671

```
[51]: clf.score(X_test, y_test)
```

```
[51]: 0.7985074626865671
```

```
[52]: print('Train Accuracy %s' % accuracy_score(y_train, y_train_pred))
print('Train F1-score %s' % f1_score(y_train, y_train_pred, average=None))
print()

print('Test Accuracy %s' % accuracy_score(y_test, y_test_pred))
print('Test F1-score %s' % f1_score(y_test, y_test_pred, average=None))
```

Train Accuracy 0.9839486356340289
 Train F1-score [0.9871134 0.9787234]

Test Accuracy 0.7985074626865671
 Test F1-score [0.84393064 0.71578947]

```
[53]: print(classification_report(y_test, y_test_pred))
```

	precision	recall	f1-score	support
0	0.81	0.88	0.84	165
1	0.78	0.66	0.72	103
accuracy			0.80	268
macro avg	0.79	0.77	0.78	268
weighted avg	0.80	0.80	0.79	268

```
[54]: results = pd.DataFrame(random_search.cv_results_)
```

```
[55]: results
```

```
[55]:
```

	mean_fit_time	std_fit_time	mean_score_time	std_score_time	\
0	0.002263	0.000234	0.001173	0.000163	
1	0.001996	0.000159	0.001081	0.000175	
2	0.002199	0.000431	0.001174	0.000261	
3	0.001748	0.000163	0.001115	0.000155	
4	0.001905	0.000201	0.001083	0.000187	
..	...	...	...	...	
195	0.001808	0.000142	0.001021	0.000096	
196	0.001453	0.000134	0.001007	0.000104	
197	0.001852	0.000125	0.000991	0.000103	
198	0.001599	0.000113	0.000963	0.000043	
199	0.001799	0.000101	0.000997	0.000074	

	param_min_samples_split	param_min_samples_leaf	param_max_depth	\
0	50	1	17	
1	5	20	12	
2	30	20	17	
3	2	1	2	
4	5	30	17	
..	...	...	...	
195	30	30	7	
196	5	100	16	
197	30	20	None	
198	100	50	13	
199	20	20	5	

	param_criterion	params	\
0	gini	{'min_samples_split': 50, 'min_samples_leaf': ...	
1	entropy	{'min_samples_split': 5, 'min_samples_leaf': 2...	
2	entropy	{'min_samples_split': 30, 'min_samples_leaf': ...	
3	entropy	{'min_samples_split': 2, 'min_samples_leaf': 1...	


```

4          entropy {'min_samples_split': 5, 'min_samples_leaf': 3...
..          ...
195        entropy {'min_samples_split': 30, 'min_samples_leaf': ...
196          gini {'min_samples_split': 5, 'min_samples_leaf': 1...
197        entropy {'min_samples_split': 30, 'min_samples_leaf': ...
198        entropy {'min_samples_split': 100, 'min_samples_leaf':...
199        entropy {'min_samples_split': 20, 'min_samples_leaf': ...

```

	split0_test_score	...	split43_test_score	split44_test_score	\
0	0.800	...	0.782258	0.741935	
1	0.848	...	0.782258	0.806452	
2	0.848	...	0.782258	0.806452	
3	0.768	...	0.782258	0.782258	
4	0.800	...	0.790323	0.766129	
..	...	...	...	...	
195	0.800	...	0.790323	0.766129	
196	0.760	...	0.790323	0.790323	
197	0.848	...	0.782258	0.806452	
198	0.752	...	0.798387	0.782258	
199	0.848	...	0.782258	0.806452	

	split45_test_score	split46_test_score	split47_test_score	\
0	0.760	0.784	0.752	
1	0.848	0.776	0.752	
2	0.848	0.768	0.752	
3	0.816	0.760	0.752	
4	0.816	0.776	0.752	
..	...	...	...	
195	0.816	0.776	0.752	
196	0.808	0.752	0.784	
197	0.848	0.768	0.752	
198	0.816	0.776	0.752	
199	0.848	0.768	0.752	

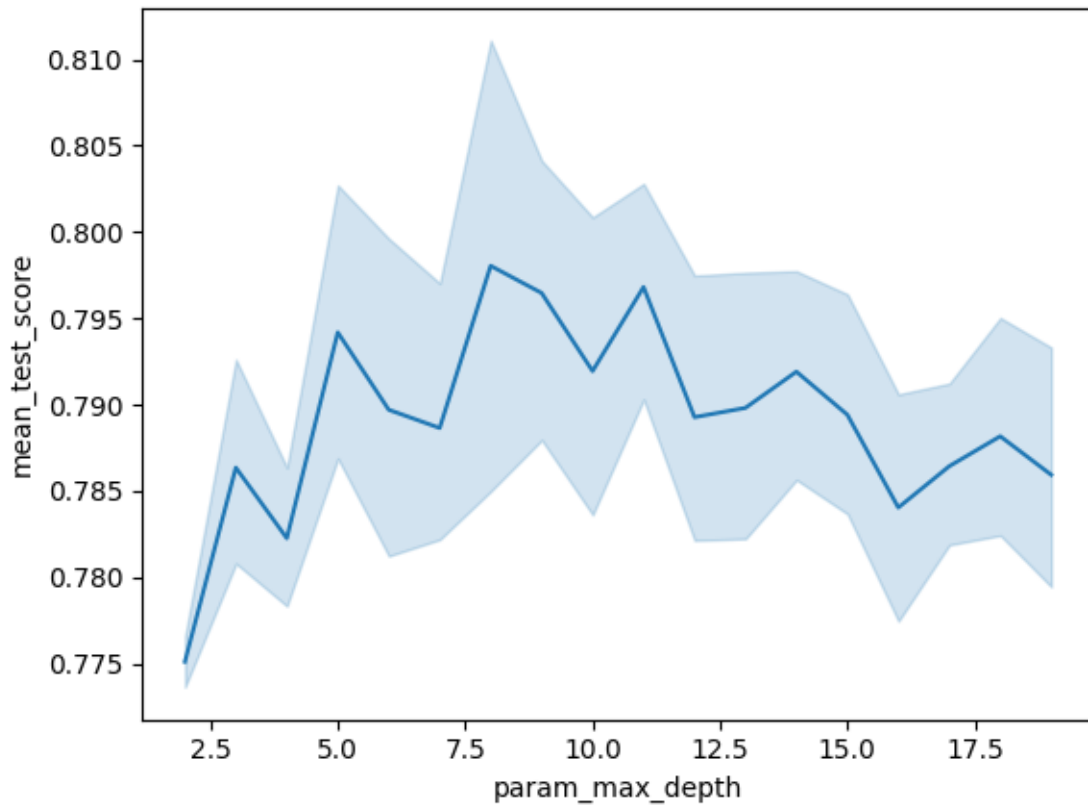
	split48_test_score	split49_test_score	mean_test_score	std_test_score	\
0	0.782258	0.854839	0.784262	0.035370	
1	0.758065	0.838710	0.793409	0.033424	
2	0.758065	0.838710	0.792769	0.033711	
3	0.782258	0.798387	0.774338	0.026770	
4	0.750000	0.806452	0.783330	0.029422	
..	...	...	...	...	
195	0.750000	0.806452	0.783650	0.029859	
196	0.766129	0.790323	0.780124	0.027891	
197	0.758065	0.838710	0.793090	0.033427	
198	0.782258	0.798387	0.775634	0.025992	
199	0.758065	0.838710	0.792609	0.033885	

	rank_test_score
0	102
1	56
2	62
3	194
4	110
..	...
195	108
196	123
197	58
198	161
199	64

[200 rows x 62 columns]

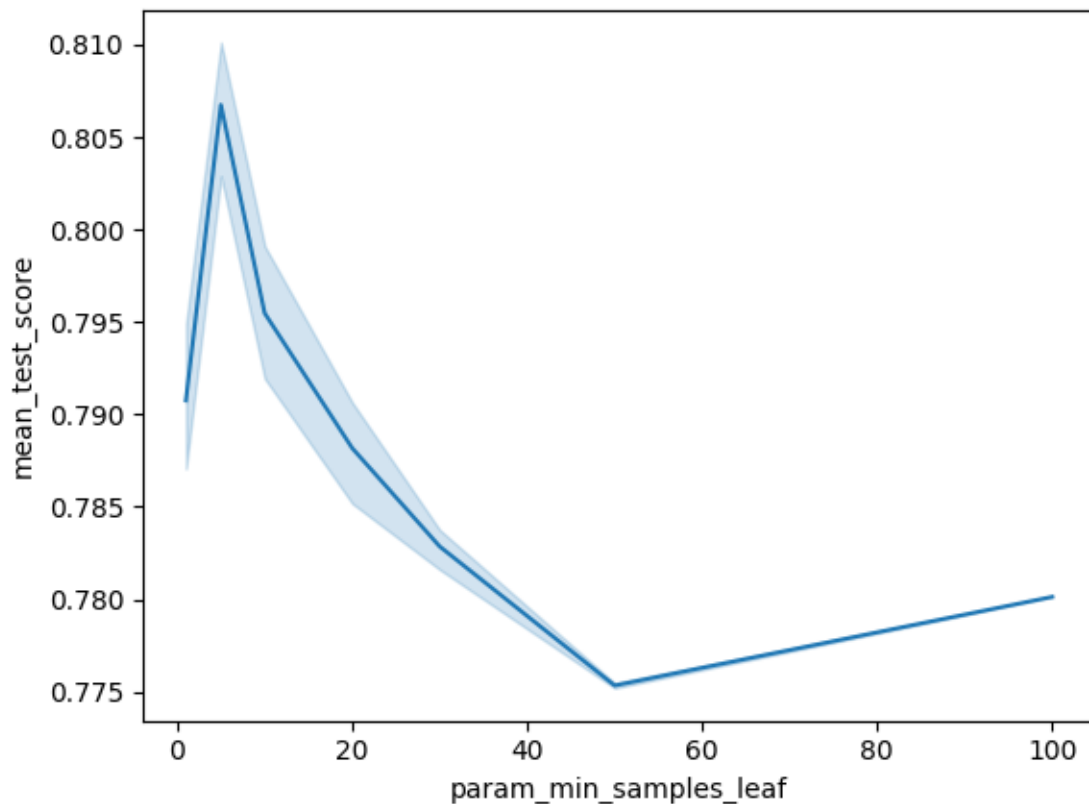
```
[56]: sns.lineplot(data=results, x="param_max_depth", y="mean_test_score")
```

```
[56]: <Axes: xlabel='param_max_depth', ylabel='mean_test_score'>
```



```
[57]: sns.lineplot(data=results, x="param_min_samples_leaf", y="mean_test_score")
```

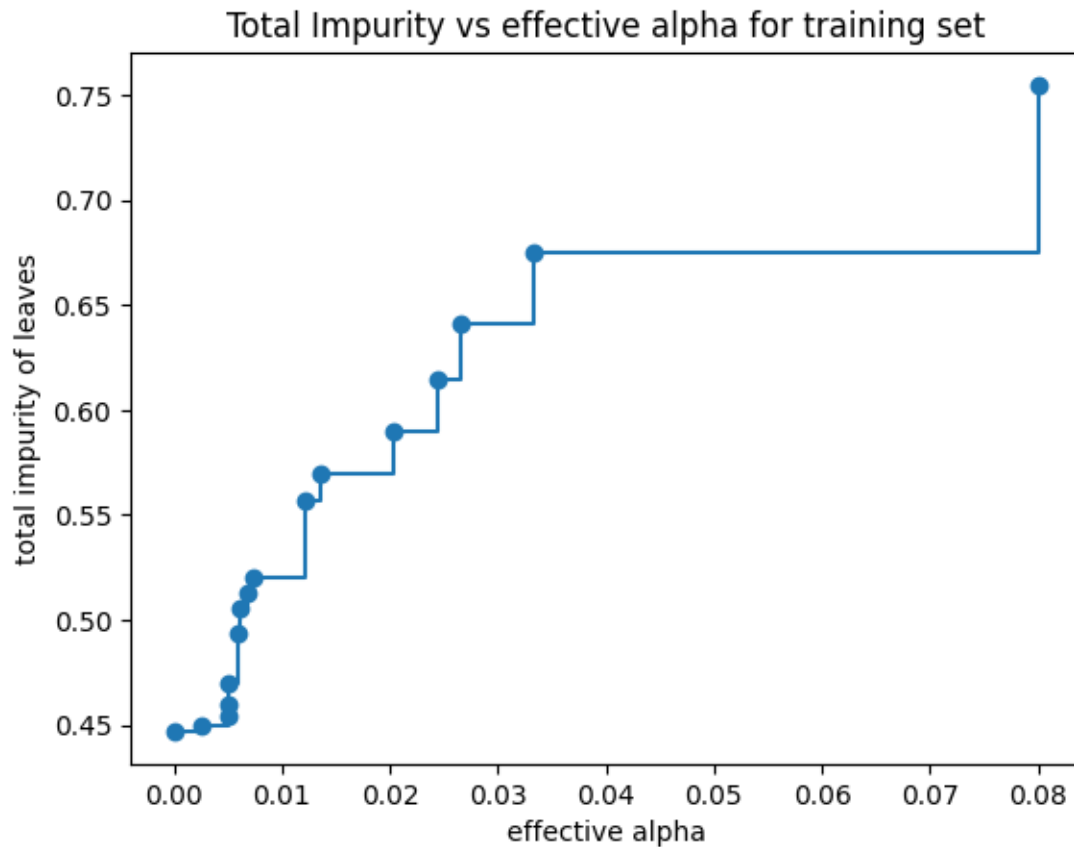
```
[57]: <Axes: xlabel='param_min_samples_leaf', ylabel='mean_test_score'>
```



```
[58]: zipped = zip(attributes, clf.feature_importances_)
zipped = sorted(zipped, key=lambda x: x[1], reverse=True)
for col, imp in zipped:
    print(col, imp)
```

```
Sex 0.40072068942597955
Pclass 0.17751028393940269
Age 0.16415838878707298
Fare 0.1566477687838109
FamilySize 0.09341600971472502
Embarked_S 0.0075468593490089
IsAlone 0.0
Embarked_C 0.0
Embarked_Q 0.0
```

```
[59]: plt.figure(figsize=(20, 4), dpi=300)
plot_tree(clf, feature_names=attributes, filled=True)
plt.show()
```

```
[63]: random_search.best_params_
```

```
[63]: {'min_samples_split': 10,
      'min_samples_leaf': 5,
      'max_depth': 6,
      'criterion': 'entropy'}
```

```
[64]: clfs = []
      for ccp_alpha in ccp_alphas:
          clf = DecisionTreeClassifier(random_state=0, ccp_alpha=ccp_alpha,
          ↳**random_search.best_params_)
          clf.fit(X_train, y_train)
          clfs.append(clf)

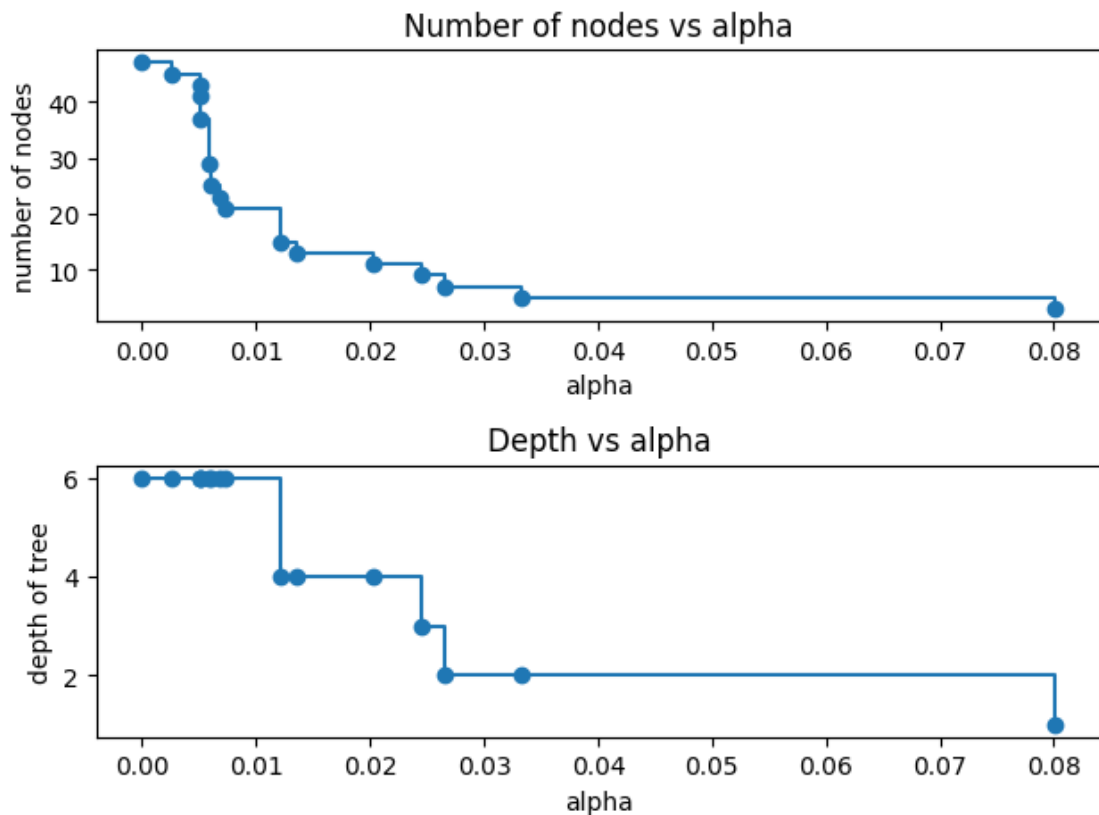
      print("Number of nodes in the last tree is: {} with ccp_alpha: {}".format(
          clfs[-1].tree_.node_count, ccp_alphas[-1]
      ))
```

Number of nodes in the last tree is: 1 with ccp_alpha: 0.20581849199058122

```
[65]: clfs = clfs[:-1]
      ccp_alphas = ccp_alphas[:-1]

      node_counts = [clf.tree_.node_count for clf in clfs]
      depth = [clf.tree_.max_depth for clf in clfs]

      fig, ax = plt.subplots(2, 1)
      ax[0].plot(ccp_alphas, node_counts, marker="o", drawstyle="steps-post")
      ax[0].set_xlabel("alpha")
      ax[0].set_ylabel("number of nodes")
      ax[0].set_title("Number of nodes vs alpha")
      ax[1].plot(ccp_alphas, depth, marker="o", drawstyle="steps-post")
      ax[1].set_xlabel("alpha")
      ax[1].set_ylabel("depth of tree")
      ax[1].set_title("Depth vs alpha")
      fig.tight_layout()
```



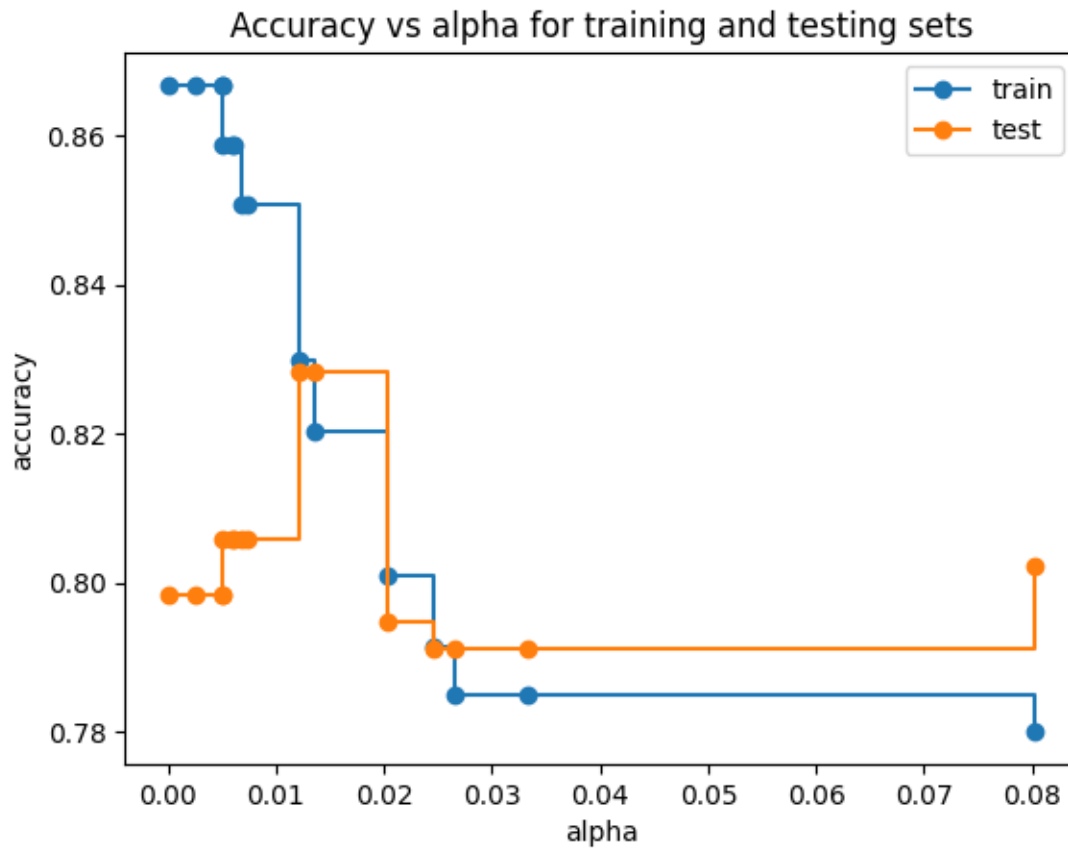
```
[66]: train_scores = [clf.score(X_train, y_train) for clf in clfs]
      test_scores = [clf.score(X_test, y_test) for clf in clfs]

      fig, ax = plt.subplots()
```

```

ax.set_xlabel("alpha")
ax.set_ylabel("accuracy")
ax.set_title("Accuracy vs alpha for training and testing sets")
ax.plot(ccp_alphas, train_scores, marker="o", label="train",
        drawstyle="steps-post")
ax.plot(ccp_alphas, test_scores, marker="o", label="test",
        drawstyle="steps-post")
ax.legend()
plt.show()

```



```

[67]: dtp = DecisionTreeClassifier()
      dtp.fit(X_train, y_train)

```

```

[67]: DecisionTreeClassifier()

```

```

[68]: y_test_pred = dtp.predict(X_test)

```

```

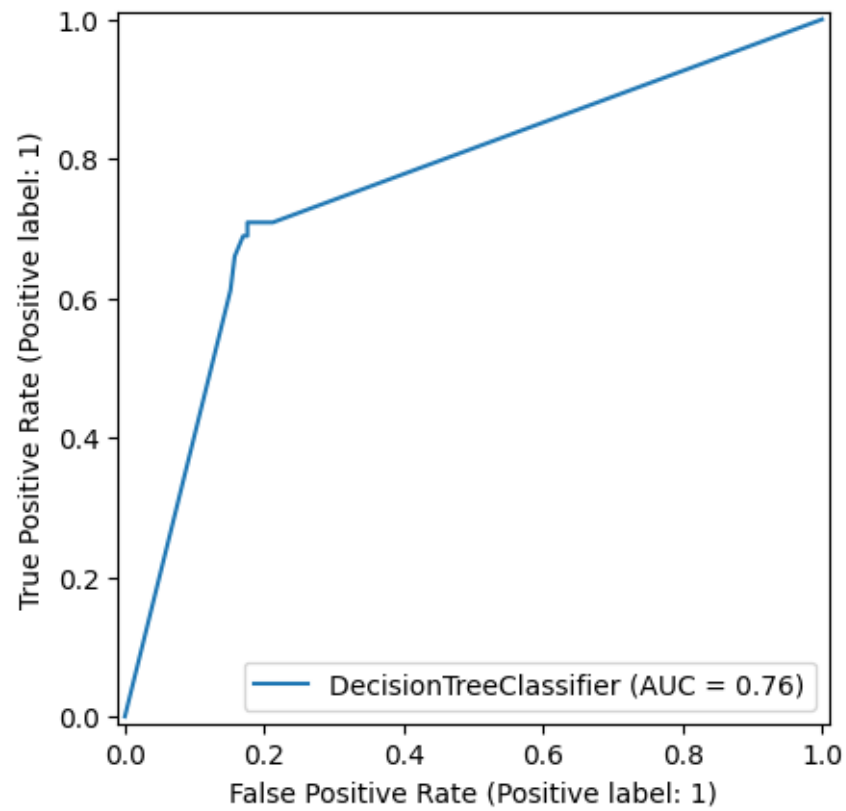
[69]: print('Train Accuracy %s' % accuracy_score(y_train, y_train_pred))
      print('Train F1-score %s' % f1_score(y_train, y_train_pred, average=None))

```

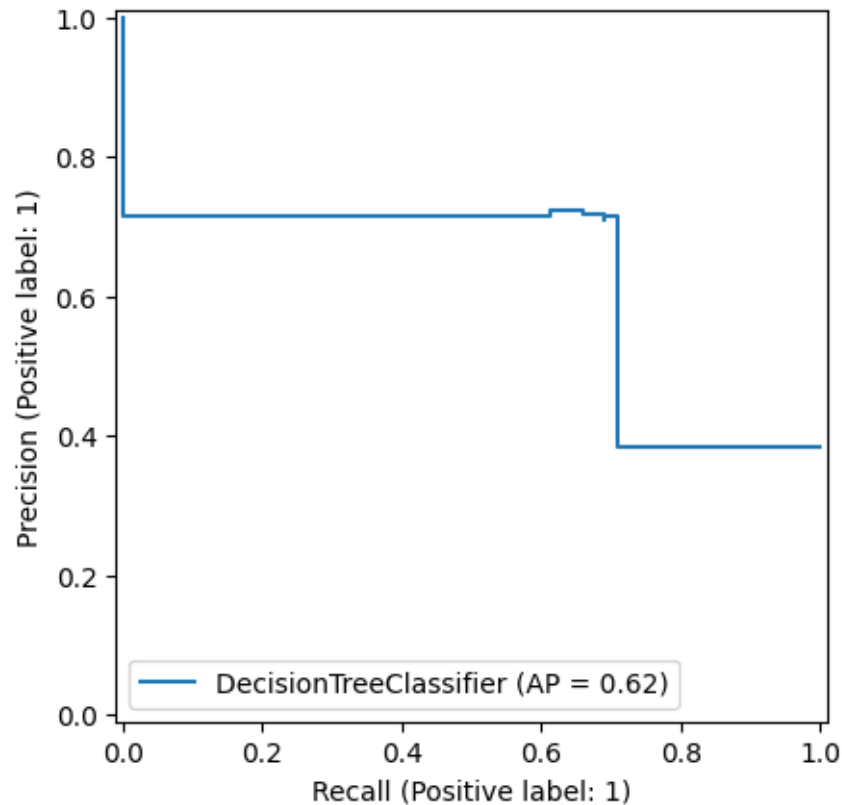

0.0.5 ROC Curves

```
[73]: from sklearn.metrics import RocCurveDisplay, PrecisionRecallDisplay
```

```
[74]: RocCurveDisplay.from_estimator(dtp, X_test, y_test)  
plt.show()
```



```
[75]: PrecisionRecallDisplay.from_estimator(dtp, X_test, y_test)  
plt.show()
```



0.0.6 Pruning Utils

In scikit-learn's decision trees, it can sometimes occur that a node split creates branches where all subsequent splits lead to the same predicted label as the node itself. In such cases, these branches do not contribute additional information to the predictions. Tree pruning functions are designed to simplify the tree by removing such redundant branches, keeping only the root node of the branch. This process reduces the complexity of the tree and is particularly useful for improving interpretability and visualization.

```
[76]: from sklearn.tree._tree import TREE_LEAF
```

```
[77]: def is_leaf(inner_tree, index):
    # Check whether node is leaf node
    return (inner_tree.children_left[index] == TREE_LEAF and
            inner_tree.children_right[index] == TREE_LEAF)

def prune_index(inner_tree, decisions, index=0):
    # Start pruning from the bottom - if we start from the top, we might miss
    # nodes that become leaves during pruning.
    # Do not use this directly - use prune_duplicate_leaves instead.
```

```

if not is_leaf(inner_tree, inner_tree.children_left[index]):
    prune_index(inner_tree, decisions, inner_tree.children_left[index])
if not is_leaf(inner_tree, inner_tree.children_right[index]):
    prune_index(inner_tree, decisions, inner_tree.children_right[index])

# Prune children if both children are leaves now and make the same decision:
if (is_leaf(inner_tree, inner_tree.children_left[index]) and
    is_leaf(inner_tree, inner_tree.children_right[index]) and
    (decisions[index] == decisions[inner_tree.children_left[index]]) and
    (decisions[index] == decisions[inner_tree.children_right[index]])):
    # turn node into a leaf by "unlinking" its children
    inner_tree.children_left[index] = TREE_LEAF
    inner_tree.children_right[index] = TREE_LEAF
    # print("Pruned {}".format(index))

def prune_duplicate_leaves(dt):
    # Remove leaves if both
    decisions = dt.tree_.value.argmax(axis=2).flatten().tolist() # Decision
    ↪ for each node
    prune_index(dt.tree_, decisions)

```

```

[78]: dt = DecisionTreeClassifier(criterion='entropy', max_depth=None,
    ↪ min_samples_split=5, min_samples_leaf=5, random_state=94)

```

```

[79]: dt.fit(X_train, y_train)

```

```

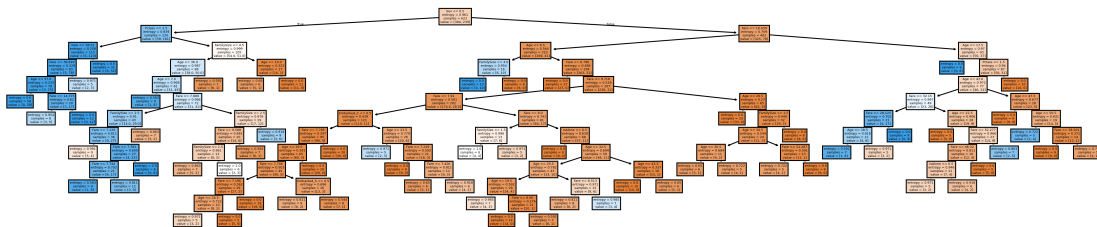
[79]: DecisionTreeClassifier(criterion='entropy', min_samples_leaf=5,
    min_samples_split=5, random_state=94)

```

```

[80]: plt.figure(figsize=(20, 4), dpi=300)
    plot_tree(dt, feature_names=attributes, filled=True)
    plt.show()

```



```

[81]: prune_duplicate_leaves(dt)

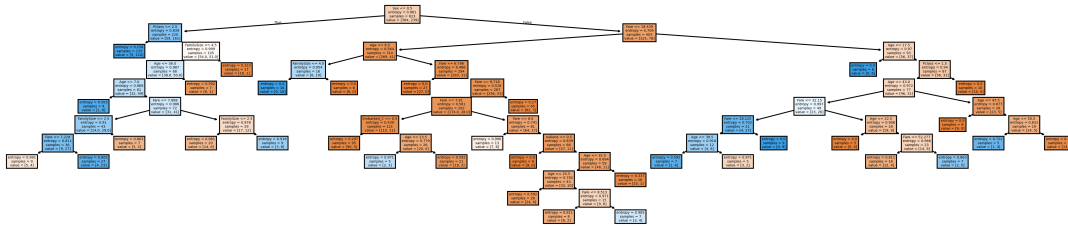
```

```

[82]: # let us visualise it again after pruning

```

```
[83]: plt.figure(figsize=(20, 4), dpi=300)
      plot_tree(dt, feature_names=attributes, filled=True)
      plt.show()
```



[83]:

[83]:

[83]:

[83]:

0.0.7 RuleTree

RuleTree is a sklearn-inspired Decision Tree library developed internally in the KDD lab. It is capable of directly handling the “children with same classes” problem directly in its parameters.

Feel free to explore RuleTrees: using scikit learn implementation is mandatory, while using RuleTree is an additional exploration you can do

For code examples on classification and code implementation, check:

<https://github.com/fismimosa/RuleTree>

https://github.com/fismimosa/RuleTree/blob/main/Examples/RuleTreeClassifier_example.ipynb

The correct import is **RuleTree.tree.RuleTreeClassifier**, not ruletree.tree.RuleTreeClassifier as in the repository, which still need to be updated

```
[84]: # Force re-installation of numpy, scipy, and RuleTree to ensure binary
      ↪compatibility
      !pip install --upgrade --force-reinstall numpy scipy RuleTree
      # IMPORTANT: Restart the runtime after this installation for changes to take
      ↪effect.
```

Collecting numpy

Using cached

numpy-2.3.5-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata
(62 kB)

Collecting scipy

```

Using cached
scipy-1.16.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata
(62 kB)
Collecting RuleTree
  Using cached RuleTree-0.0.4.post2-py3-none-any.whl
Collecting numpy
  Using cached
numpy-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(61 kB)
Collecting scikit-learn>=1.5.0 (from RuleTree)
  Using cached scikit_learn-1.7.2-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (11 kB)
Collecting pandas (from RuleTree)
  Using cached pandas-2.3.3-cp312-cp312-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (91 kB)
Collecting category-encoders (from RuleTree)
  Using cached category_encoders-2.9.0-py3-none-any.whl.metadata (7.9 kB)
Collecting threadpoolctl (from RuleTree)
  Using cached threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Collecting tqdm (from RuleTree)
  Using cached tqdm-4.67.1-py3-none-any.whl.metadata (57 kB)
Collecting progress-table (from RuleTree)
  Using cached progress_table-3.1.2-py3-none-any.whl.metadata (5.7 kB)
Collecting graphviz>=0.20.3 (from RuleTree)
  Using cached graphviz-0.21-py3-none-any.whl.metadata (12 kB)
Collecting numba (from RuleTree)
  Using cached
numba-0.62.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata
(2.8 kB)
Collecting psutil (from RuleTree)
  Using cached psutil-7.1.3-cp36-abi3-
manylinux2010_x86_64.manylinux_2_12_x86_64.manylinux_2_28_x86_64.whl.metadata
(23 kB)
Collecting setuptools (from RuleTree)
  Using cached setuptools-80.9.0-py3-none-any.whl.metadata (6.6 kB)
Collecting matplotlib (from RuleTree)
  Using cached matplotlib-3.10.7-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (11 kB)
Collecting tempfile312 (from RuleTree)
  Using cached tempfile312-1.0.1-py3-none-any.whl.metadata (1.7 kB)
Collecting joblib>=1.2.0 (from scikit-learn>=1.5.0->RuleTree)
  Using cached joblib-1.5.2-py3-none-any.whl.metadata (5.6 kB)
Collecting patsy>=0.5.1 (from category-encoders->RuleTree)
  Using cached patsy-1.0.2-py2.py3-none-any.whl.metadata (3.6 kB)
Collecting statsmodels>=0.9.0 (from category-encoders->RuleTree)
  Using cached statsmodels-0.14.5-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata
(9.5 kB)

```

Collecting python-dateutil>=2.8.2 (from pandas->RuleTree)
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting pytz>=2020.1 (from pandas->RuleTree)
Using cached pytz-2025.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas->RuleTree)
Using cached tzdata-2025.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib->RuleTree)
Using cached contourpy-1.3.3-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib->RuleTree)
Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib->RuleTree)
Using cached fonttools-4.60.1-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl.metadata (112 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib->RuleTree)
Using cached kiwisolver-1.4.9-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (6.3 kB)
Collecting packaging>=20.0 (from matplotlib->RuleTree)
Using cached packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pillow>=8 (from matplotlib->RuleTree)
Using cached pillow-12.0.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (8.8 kB)
Collecting pyparsing>=3 (from matplotlib->RuleTree)
Using cached pyparsing-3.2.5-py3-none-any.whl.metadata (5.0 kB)
Collecting llvmlite<0.46,>=0.45.0dev0 (from numba->RuleTree)
Using cached llvmlite-0.45.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (4.9 kB)
Collecting colorama (from progress-table->RuleTree)
Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas->RuleTree)
Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Using cached
scipy-1.16.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (35.7 MB)
Using cached
numpy-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.0 MB)
Using cached graphviz-0.21-py3-none-any.whl (47 kB)
Using cached
scikit_learn-1.7.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (9.5 MB)
Using cached threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Using cached category_encoders-2.9.0-py3-none-any.whl (85 kB)
Using cached
pandas-2.3.3-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (12.4 MB)

```

Using cached
matplotlib-3.10.7-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
(8.7 MB)
Using cached
numba-0.62.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (3.8 MB)
Using cached progress_table-3.1.2-py3-none-any.whl (20 kB)
Using cached psutil-7.1.3-cp36-abi3-
manylinux2010_x86_64.manylinux_2_12_x86_64.manylinux_2_28_x86_64.whl (263 kB)
Using cached setuptools-80.9.0-py3-none-any.whl (1.2 MB)
Using cached tempfile312-1.0.1-py3-none-any.whl (19 kB)
Using cached tqdm-4.67.1-py3-none-any.whl (78 kB)
Using cached
contourpy-1.3.3-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (362
kB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Using cached fonttools-4.60.1-cp312-cp312-
manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_6
4.whl (4.9 MB)
Using cached joblib-1.5.2-py3-none-any.whl (308 kB)
Using cached
kiwisolver-1.4.9-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.5
MB)
Using cached
llvmlite-0.45.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (56.3
MB)
Using cached packaging-25.0-py3-none-any.whl (66 kB)
Using cached patsy-1.0.2-py2.py3-none-any.whl (233 kB)
Using cached
pillow-12.0.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (7.0
MB)
Using cached pyparsing-3.2.5-py3-none-any.whl (113 kB)
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached pytz-2025.2-py2.py3-none-any.whl (509 kB)
Using cached statsmodels-0.14.5-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (10.4 MB)
Using cached tzdata-2025.2-py2.py3-none-any.whl (347 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: pytz, tzdata, tqdm, threadpoolctl, tempfile312,
six, setuptools, pyparsing, psutil, pillow, packaging, numpy, llvmlite,
kiwisolver, joblib, graphviz, fonttools, cycler, colorama, scipy, python-
dateutil, progress-table, patsy, numba, contourpy, scikit-learn, pandas,
matplotlib, statsmodels, category-encoders, RuleTree
  Attempting uninstall: pytz
    Found existing installation: pytz 2025.2
    Uninstalling pytz-2025.2:
      Successfully uninstalled pytz-2025.2
  Attempting uninstall: tzdata

```

Found existing installation: tzdata 2025.2
Uninstalling tzdata-2025.2:
 Successfully uninstalled tzdata-2025.2
Attempting uninstall: tqdm
 Found existing installation: tqdm 4.67.1
 Uninstalling tqdm-4.67.1:
 Successfully uninstalled tqdm-4.67.1
Attempting uninstall: threadpoolctl
 Found existing installation: threadpoolctl 3.6.0
 Uninstalling threadpoolctl-3.6.0:
 Successfully uninstalled threadpoolctl-3.6.0
Attempting uninstall: tempfile312
 Found existing installation: tempfile312 1.0.1
 Uninstalling tempfile312-1.0.1:
 Successfully uninstalled tempfile312-1.0.1
Attempting uninstall: six
 Found existing installation: six 1.17.0
 Uninstalling six-1.17.0:
 Successfully uninstalled six-1.17.0
Attempting uninstall: setuptools
 Found existing installation: setuptools 80.9.0
 Uninstalling setuptools-80.9.0:
 Successfully uninstalled setuptools-80.9.0
Attempting uninstall: pyparsing
 Found existing installation: pyparsing 3.2.5
 Uninstalling pyparsing-3.2.5:
 Successfully uninstalled pyparsing-3.2.5
Attempting uninstall: psutil
 Found existing installation: psutil 7.1.3
 Uninstalling psutil-7.1.3:
 Successfully uninstalled psutil-7.1.3
Attempting uninstall: pillow
 Found existing installation: pillow 12.0.0
 Uninstalling pillow-12.0.0:
 Successfully uninstalled pillow-12.0.0
Attempting uninstall: packaging
 Found existing installation: packaging 25.0
 Uninstalling packaging-25.0:
 Successfully uninstalled packaging-25.0
Attempting uninstall: numpy
 Found existing installation: numpy 1.26.4
 Uninstalling numpy-1.26.4:
 Successfully uninstalled numpy-1.26.4
Attempting uninstall: llvmlite
 Found existing installation: llvmlite 0.45.1
 Uninstalling llvmlite-0.45.1:
 Successfully uninstalled llvmlite-0.45.1
Attempting uninstall: kiwisolver

Found existing installation: kiwisolver 1.4.9
Uninstalling kiwisolver-1.4.9:
 Successfully uninstalled kiwisolver-1.4.9
Attempting uninstall: joblib
 Found existing installation: joblib 1.5.2
 Uninstalling joblib-1.5.2:
 Successfully uninstalled joblib-1.5.2
Attempting uninstall: graphviz
 Found existing installation: graphviz 0.21
 Uninstalling graphviz-0.21:
 Successfully uninstalled graphviz-0.21
Attempting uninstall: fonttools
 Found existing installation: fonttools 4.60.1
 Uninstalling fonttools-4.60.1:
 Successfully uninstalled fonttools-4.60.1
Attempting uninstall: cyclcr
 Found existing installation: cyclcr 0.12.1
 Uninstalling cyclcr-0.12.1:
 Successfully uninstalled cyclcr-0.12.1
Attempting uninstall: colorama
 Found existing installation: colorama 0.4.6
 Uninstalling colorama-0.4.6:
 Successfully uninstalled colorama-0.4.6
Attempting uninstall: scipy
 Found existing installation: scipy 1.16.3
 Uninstalling scipy-1.16.3:
 Successfully uninstalled scipy-1.16.3
Attempting uninstall: python-dateutil
 Found existing installation: python-dateutil 2.9.0.post0
 Uninstalling python-dateutil-2.9.0.post0:
 Successfully uninstalled python-dateutil-2.9.0.post0
Attempting uninstall: progress-table
 Found existing installation: progress-table 3.1.2
 Uninstalling progress-table-3.1.2:
 Successfully uninstalled progress-table-3.1.2
Attempting uninstall: patsy
 Found existing installation: patsy 1.0.2
 Uninstalling patsy-1.0.2:
 Successfully uninstalled patsy-1.0.2
Attempting uninstall: numba
 Found existing installation: numba 0.62.1
 Uninstalling numba-0.62.1:
 Successfully uninstalled numba-0.62.1
Attempting uninstall: contourpy
 Found existing installation: contourpy 1.3.3
 Uninstalling contourpy-1.3.3:
 Successfully uninstalled contourpy-1.3.3
Attempting uninstall: scikit-learn

```
Found existing installation: scikit-learn 1.7.2
Uninstalling scikit-learn-1.7.2:
  Successfully uninstalled scikit-learn-1.7.2
Attempting uninstall: pandas
Found existing installation: pandas 2.3.3
Uninstalling pandas-2.3.3:
  Successfully uninstalled pandas-2.3.3
Attempting uninstall: matplotlib
Found existing installation: matplotlib 3.10.7
Uninstalling matplotlib-3.10.7:
  Successfully uninstalled matplotlib-3.10.7
Attempting uninstall: statsmodels
Found existing installation: statsmodels 0.14.5
Uninstalling statsmodels-0.14.5:
  Successfully uninstalled statsmodels-0.14.5
Attempting uninstall: category-encoders
Found existing installation: category_encoders 2.9.0
Uninstalling category_encoders-2.9.0:
  Successfully uninstalled category_encoders-2.9.0
Attempting uninstall: RuleTree
Found existing installation: RuleTree 0.0.4.post2
Uninstalling RuleTree-0.0.4.post2:
  Successfully uninstalled RuleTree-0.0.4.post2
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

ipython 7.34.0 requires jedi>=0.16, which is not installed.

google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.3.3 which is incompatible.

opencv-python 4.12.0.88 requires numpy<2.3.0,>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.

jax 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.

opencv-contrib-python 4.12.0.88 requires numpy<2.3.0,>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.

pytensor 2.35.1 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.

jaxlib 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.

cuml-cu12 25.10.0 requires numba<0.62.0a0,>=0.60.0, but you have numba 0.62.1 which is incompatible.

opencv-python-headless 4.12.0.88 requires numpy<2.3.0,>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.

shap 0.50.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.

cudf-cu12 25.10.0 requires numba<0.62.0a0,>=0.60.0, but you have numba 0.62.1 which is incompatible.

gradio 5.49.1 requires pillow<12.0,>=8.0, but you have pillow 12.0.0 which is incompatible.

Successfully installed RuleTree-0.0.4.post2 category-encoders-2.9.0
colorama-0.4.6 contourpy-1.3.3 cycler-0.12.1 fonttools-4.60.1 graphviz-0.21
joblib-1.5.2 kiwisolver-1.4.9 llvmlite-0.45.1 matplotlib-3.10.7 numba-0.62.1
numpy-1.26.4 packaging-25.0 pandas-2.3.3 patsy-1.0.2 pillow-12.0.0 progress-
table-3.1.2 psutil-7.1.3 pyparsing-3.2.5 python-dateutil-2.9.0.post0 pytz-2025.2
scikit-learn-1.7.2 scipy-1.16.3 setuptools-80.9.0 six-1.17.0 statsmodels-0.14.5
tempfile312-1.0.1 threadpoolctl-3.6.0 tqdm-4.67.1 tzdata-2025.2

[100]: *#Initialize the RuleTreeClassifier and train it on your dataset*

```

from RuleTree.tree.RuleTreeClassifier import RuleTreeClassifier
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

```

```
[106]: X_train.shape, attributes
```

```

[106]: ((623, 9),
        ['Pclass',
         'Sex',
         'Age',
         'Fare',
         'FamilySize',
         'IsAlone',
         'Embarked_C',
         'Embarked_Q',
         'Embarked_S'])

```

```

[107]: rtc = RuleTreeClassifier(
        max_depth=3,
        criterion='gini',
        prune_useless_leaves=True, ##prune_useless_leaves performing the pruning
        ↪ internally for you without the need to access external functions
        random_state=42
    )

```

IMPORTANT DETAIL: for training, both X_train and y_train should be numpy objects, not pandas series

```

[108]: # Fit the RuleTreeClassifier to the training data
        rtc.fit(X_train, np.array(y_train))

```

```

[108]: RuleTreeClassifier(base_stumps=[(1.0,
                                         DecisionTreeStumpClassifier(ccp_alpha=0.0,
                                                                       class_weight=None,
                                                                       criterion='gini',
                                                                       max_depth=1,
                                                                       max_features=None,
                                                                       min_impurity_decrease=0.0,
                                                                       min_samples_leaf=1,
                                                                       min_samples_split=2,
                                                                       min_weight_fraction_leaf=0.0,
                                                                       monotonic_cst=None,
                                                                       random_state=42,
                                                                       splitter='best'))],
                           max_depth=3, prune_useless_leaves=True, random_state=42)

```

```
[109]: # Extract the rules as a dictionary using feature names
rules = rtc.get_rules(columns_names=attributes)
```

```
[110]: # Print textual rules of trained tree
rtc.print_rules(rules, columns_names=attributes)
```

```
|--- Sex <= 0.5 623
|   |--- Pclass <= 2.5 220
|   |   |--- Fare <= 5.75 115
|   |   |   output: 0
|   |   |--- Fare > 5.75
|   |   |   output: 1
|   |--- Pclass > 2.5
|   |   |--- FamilySize <= 4.5 105
|   |   |   output: 1
|   |   |--- FamilySize > 4.5
|   |   |   output: 0
|--- Sex > 0.5
|   |--- Age <= 6.5 403
|   |   |--- FamilySize <= 4.5 20
|   |   |   output: 1
|   |   |--- FamilySize > 4.5
|   |   |   output: 0
|   |--- Age > 6.5
|   |   output: 0
```

```
[111]: y_pred = rtc.predict(X_test)
```

```
[112]: print('Accuracy result\n', accuracy_score(y_test, y_pred))
```

```
Accuracy result
0.8283582089552238
```

```
[113]: # Obtain global feature importances similar to sklearn feature_importances_of_
      ↪ a trained Tree
feature_importances = rtc.compute_feature_importances()
for attr, imp in zip(attributes, feature_importances):
    print(attr, imp)
```

```
Pclass 0.17992977235110494
Sex 0.6098058869982659
Age 0.07982962656091805
Fare 0.01365006097271372
FamilySize 0.11678465311699747
IsAlone 0.0
Embarked_C 0.0
Embarked_Q 0.0
Embarked_S 0.0
```

```
[114]: # Get local feature importances similar to https://github.com/andosa/
        ↪treeinterpreter implementation
        # Prediction is the sum of bias and feature contributions
        prediction, bias, contributions = rtc.local_interpretation(X_test,
        ↪joint_contribution= True)
```

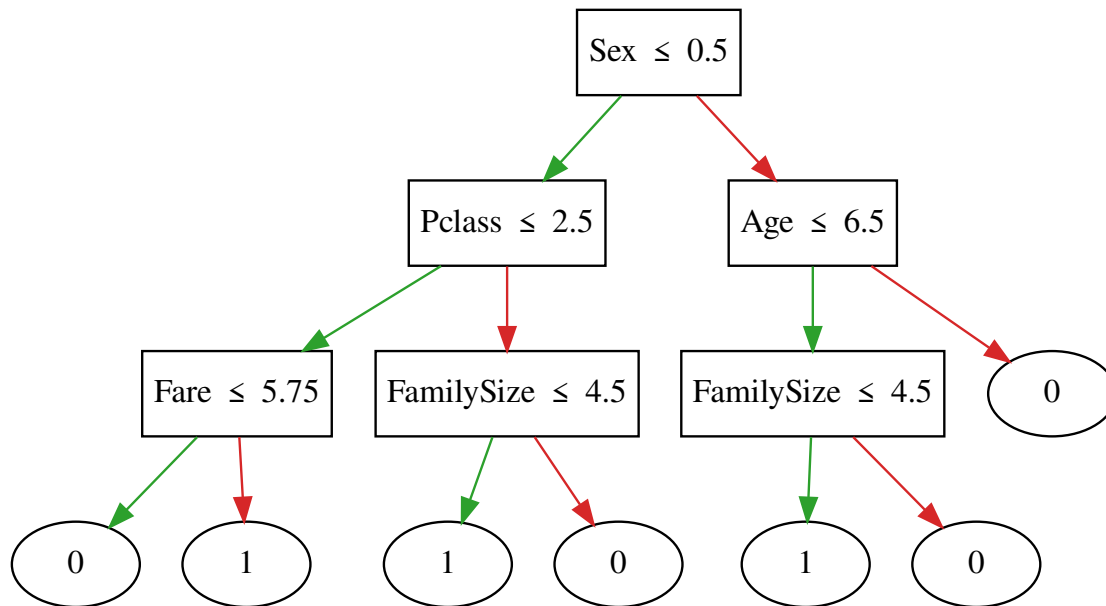
```
[115]: prediction[0], bias[0], contributions[0]
```

```
[115]: (array([0.83289817, 0.16710183]),
        array([0.61637239, 0.38362761]),
        {(1,): array([ 0.19007922, -0.19007922]),
         (1, 2): array([ 0.02644656, -0.02644656])})
```

```
[115]:
```

```
[116]: rtc.export_graphviz(columns_names = attributes)
```

```
[116]:
```



```
[117]: rtc.export_graphviz(columns_names = attributes, filename = 'data/
        ↪rtc_plot_tree') #adding filename will save a pdf of the plot
```

```
[118]: #directly obtain information on the nodes in the tree

        rtc.root.node_to_dict()
```

```
[118]: {'node_id': 'R',
        'is_leaf': False,
        'prediction': 0,
```

```

'prediction_probability': [0.6163723916532905, 0.38362760834670945],
'prediction_classes_': [0, 1],
'features': 9,
'left_node': 'Rl',
'right_node': 'Rr',
'feature_idx': 1,
'threshold': 0.5,
'is_categorical': None,
'samples': 623,
'feature_name': 'X_1',
'textual_rule': 'X_1 <= 0.5\t623',
'blob_rule': 'X_1 <= 0.5',
'graphviz_rule': {'label': 'X_1 0.5'},
'not_textual_rule': 'X_1 > 0.5',
'not_blob_rule': 'X_1 > 0.5',
'not_graphviz_rule': {'label': 'X_1 > 0.5'},
'stump_type': 'RuleTree.stumps.classification.DecisionTreeStumpClassifier',
'impurity': 0.4729149329217863,
'args': {'is_oblique': False,
'is_pivotal': False,
'unique_val_enum': None,
'coefficients': None,
'max_depth': 1,
'criterion': 'gini',
'splitter': 'best',
'min_samples_split': 2,
'min_samples_leaf': 1,
'min_weight_fraction_leaf': 0.0,
'max_features': None,
'random_state': 42,
'min_impurity_decrease': 0.0,
'class_weight': None,
'ccp_alpha': 0.0,
'monotonic_cst': None},
'split': {'args': {}}}

```

```

[119]: #navigate the tree from the root with .node_l and .node_r
rtc.root.node_l.node_to_dict()

```

```

[119]: {'node_id': 'Rl',
'is_leaf': False,
'prediction': 1,
'prediction_probability': [0.2681818181818182, 0.7318181818181818],
'prediction_classes_': [0, 1],
'features': 9,
'left_node': 'Rll',
'right_node': 'Rlr',

```

```

'feature_idx': 0,
'threshold': 2.5,
'is_categorical': None,
'samples': 220,
'feature_name': 'X_0',
'textual_rule': 'X_0 <= 2.5\t220',
'blob_rule': 'X_0 <= 2.5',
'graphviz_rule': {'label': 'X_0 2.5'},
'not_textual_rule': 'X_0 > 2.5',
'not_blob_rule': 'X_0 > 2.5',
'not_graphviz_rule': {'label': 'X_0 > 2.5'},
'stump_type': 'RuleTree.stumps.classification.DecisionTreeStumpClassifier',
'impurity': 0.3925206611570248,
'args': {'is_oblique': False,
'is_pivotal': False,
'unique_val_enum': None,
'coefficients': None,
'max_depth': 1,
'criterion': 'gini',
'splitter': 'best',
'min_samples_split': 2,
'min_samples_leaf': 1,
'min_weight_fraction_leaf': 0.0,
'max_features': None,
'random_state': 42,
'min_impurity_decrease': 0.0,
'class_weight': None,
'ccp_alpha': 0.0,
'monotonic_cst': None},
'split': {'args': {}}}

```

```

[121]: !pip install nbconvert
!apt-get install texlive texlive-xetex texlive-latex-extra pandoc

```

```

Requirement already satisfied: nbconvert in /usr/local/lib/python3.12/dist-
packages (7.16.6)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-
packages (from nbconvert) (4.13.5)
Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.12/dist-
packages (from bleach[css]!=5.0.0->nbconvert) (6.3.0)
Requirement already satisfied: defusedxml in /usr/local/lib/python3.12/dist-
packages (from nbconvert) (0.7.1)
Requirement already satisfied: Jinja2>=3.0 in /usr/local/lib/python3.12/dist-
packages (from nbconvert) (3.1.6)
Requirement already satisfied: jupyter-core>=4.7 in
/usr/local/lib/python3.12/dist-packages (from nbconvert) (5.9.1)
Requirement already satisfied: jupyterlab-pygments in

```


/usr/local/lib/python3.12/dist-packages (from nbconvert) (0.3.0)
 Requirement already satisfied: markupsafe>=2.0 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.0.3)
 Requirement already satisfied: mistune<4,>=2.0.3 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.1.4)
 Requirement already satisfied: nbclient>=0.5.0 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.10.2)
 Requirement already satisfied: nbformat>=5.7 in /usr/local/lib/python3.12/dist-
 packages (from nbconvert) (5.10.4)
 Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-
 packages (from nbconvert) (25.0)
 Requirement already satisfied: pandocfilters>=1.4.1 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (1.5.1)
 Requirement already satisfied: pygments>=2.4.1 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (2.19.2)
 Requirement already satisfied: traitlets>=5.1 in /usr/local/lib/python3.12/dist-
 packages (from nbconvert) (5.7.1)
 Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-
 packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert) (0.5.1)
 Requirement already satisfied: tinycss2<1.5,>=1.1.0 in
 /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->nbconvert)
 (1.4.0)
 Requirement already satisfied: platformdirs>=2.5 in
 /usr/local/lib/python3.12/dist-packages (from jupyter-core>=4.7->nbconvert)
 (4.5.0)
 Requirement already satisfied: jupyter-client>=6.1.12 in
 /usr/local/lib/python3.12/dist-packages (from nbclient>=0.5.0->nbconvert)
 (7.4.9)
 Requirement already satisfied: fastjsonschema>=2.15 in
 /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert) (2.21.2)
 Requirement already satisfied: jsonschema>=2.6 in
 /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert) (4.25.1)
 Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-
 packages (from beautifulsoup4->nbconvert) (2.8)
 Requirement already satisfied: typing-extensions>=4.0.0 in
 /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->nbconvert)
 (4.15.0)
 Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-
 packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (25.4.0)
 Requirement already satisfied: jsonschema-specifications>=2023.03.6 in
 /usr/local/lib/python3.12/dist-packages (from
 jsonschema>=2.6->nbformat>=5.7->nbconvert) (2025.9.1)
 Requirement already satisfied: referencing>=0.28.4 in
 /usr/local/lib/python3.12/dist-packages (from
 jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.37.0)
 Requirement already satisfied: rpds-py>=0.7.1 in /usr/local/lib/python3.12/dist-
 packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.29.0)
 Requirement already satisfied: entrypoints in /usr/local/lib/python3.12/dist-

packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (0.4)
 Requirement already satisfied: nest-asyncio>=1.5.4 in
 /usr/local/lib/python3.12/dist-packages (from jupyter-
 client>=6.1.12->nbclient>=0.5.0->nbconvert) (1.6.0)
 Requirement already satisfied: python-dateutil>=2.8.2 in
 /usr/local/lib/python3.12/dist-packages (from jupyter-
 client>=6.1.12->nbclient>=0.5.0->nbconvert) (2.9.0.post0)
 Requirement already satisfied: pyzmq>=23.0 in /usr/local/lib/python3.12/dist-
 packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (26.2.1)
 Requirement already satisfied: tornado>=6.2 in /usr/local/lib/python3.12/dist-
 packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (6.5.1)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
 packages (from python-dateutil>=2.8.2->jupyter-
 client>=6.1.12->nbclient>=0.5.0->nbconvert) (1.17.0)
 Reading package lists... Done
 Building dependency tree... Done
 Reading state information... Done
 pandoc is already the newest version (2.9.2.1-3ubuntu2).
 pandoc set to manually installed.

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
 fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
 libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
 libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
 lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
 ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
 teckit tex-common tex-gyre texlive-base texlive-binaries
 texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
 texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
 libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
 poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
 fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
 fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
 | postscript-viewer perl-tk xpdf | pdf-viewer xzdec
 texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
 icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
 texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
 texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
 default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
 fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
 libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
 libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13

```

lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tlutils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils
0 upgraded, 54 newly installed, 0 to remove and 41 not upgraded.
Need to get 182 MB of archives.
After this operation, 571 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-logging-
java all 1.2-2 [60.3 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libptexenc1

```

amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
 Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration
 all 1.18 [5,336 B]
 Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64
 3.0.2-7ubuntu2.11 [50.1 kB]
 Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems
 all 3.3.5-2ubuntu1.2 [228 kB]
 Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1
 [5,100 B]
 Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
 kB]
 Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
 0.1.1-2 [12.6 kB]
 Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
 all 1.7.0-3ubuntu0.2 [52.5 kB]
 Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all
 0.3.2-1ubuntu0.1 [24.9 kB]
 Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
 Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynctex2
 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
 Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
 2.5.11+ds1-1 [421 kB]
 Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
 Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2
 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzzip-0-13 amd64
 0.13.72+dfsg.1-1.1 [27.0 kB]
 Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all
 1:1.0.5-0ubuntu2 [578 kB]
 Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64
 1:7.7+6build2 [94.6 kB]
 Get:37 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all
 2.004.5-6.1 [9,471 kB]
 Get:38 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style
 all 12.2-1ubuntu1 [185 kB]
 Get:39 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 t1utils amd64
 1.41-4build2 [61.3 kB]
 Get:40 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 teckit amd64
 2.5.11+ds1-1 [699 kB]
 Get:41 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-gyre all
 20180621-3.1 [6,209 kB]
 Get:42 <http://archive.ubuntu.com/ubuntu> jammy-updates/universe amd64 texlive-
 binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
 Get:43 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-base all
 2021.20220204-1 [21.0 MB]
 Get:44 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-fonts-

```

recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 8s (22.5 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 121713 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.

```

```

Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...

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```

Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../32-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...

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```

Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.

```



```

Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr

```

```

/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

```

```

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.

```

```
[122]: !jupyter nbconvert --to pdf Classification_Decision_Tree.ipynb
```

```

[NbConvertApp] WARNING | pattern 'Classification_Decision_Tree.ipynb' matched no
files
This application is used to convert notebook files (*.ipynb)
    to various other formats.

```

```
WARNING: THE COMMANDLINE INTERFACE MAY CHANGE IN FUTURE RELEASES.
```

Options

```
=====
```

The options below are convenience aliases to configurable class-options, as listed in the "Equivalent to" description-line of the aliases.

To see all configurable class-options for some <cmd>, use:

```
<cmd> --help-all
```

--debug

set log level to logging.DEBUG (maximize logging output)

Equivalent to: [--Application.log_level=10]

--show-config

Show the application's configuration (human-readable format)

Equivalent to: [--Application.show_config=True]

--show-config-json

Show the application's configuration (json format)

Equivalent to: [--Application.show_config_json=True]

--generate-config

```

    generate default config file
    Equivalent to: [--JupyterApp.generate_config=True]
-y
    Answer yes to any questions instead of prompting.
    Equivalent to: [--JupyterApp.answer_yes=True]
--execute
    Execute the notebook prior to export.
    Equivalent to: [--ExecutePreprocessor.enabled=True]
--allow-errors
    Continue notebook execution even if one of the cells throws an error and
    include the error message in the cell output (the default behaviour is to abort
    conversion). This flag is only relevant if '--execute' was specified, too.
    Equivalent to: [--ExecutePreprocessor.allow_errors=True]
--stdin
    read a single notebook file from stdin. Write the resulting notebook with
    default basename 'notebook.*'
    Equivalent to: [--NbConvertApp.from_stdin=True]
--stdout
    Write notebook output to stdout instead of files.
    Equivalent to: [--NbConvertApp.writer_class=StdoutWriter]
--inplace
    Run nbconvert in place, overwriting the existing notebook (only
    relevant when converting to notebook format)
    Equivalent to: [--NbConvertApp.use_output_suffix=False
--NbConvertApp.export_format=notebook --FilesWriter.build_directory=]
--clear-output
    Clear output of current file and save in place,
    overwriting the existing notebook.
    Equivalent to: [--NbConvertApp.use_output_suffix=False
--NbConvertApp.export_format=notebook --FilesWriter.build_directory=
--ClearOutputPreprocessor.enabled=True]
--coalesce-streams
    Coalesce consecutive stdout and stderr outputs into one stream (within each
    cell).
    Equivalent to: [--NbConvertApp.use_output_suffix=False
--NbConvertApp.export_format=notebook --FilesWriter.build_directory=
--CoalesceStreamsPreprocessor.enabled=True]
--no-prompt
    Exclude input and output prompts from converted document.
    Equivalent to: [--TemplateExporter.exclude_input_prompt=True
--TemplateExporter.exclude_output_prompt=True]
--no-input
    Exclude input cells and output prompts from converted document.
    This mode is ideal for generating code-free reports.
    Equivalent to: [--TemplateExporter.exclude_output_prompt=True
--TemplateExporter.exclude_input=True
--TemplateExporter.exclude_input_prompt=True]
--allow-chromium-download

```

Whether to allow downloading chromium if no suitable version is found on the system.

Equivalent to: [--WebPDFExporter.allow_chromium_download=True]

--disable-chromium-sandbox

Disable chromium security sandbox when converting to PDF..

Equivalent to: [--WebPDFExporter.disable_sandbox=True]

--show-input

Shows code input. This flag is only useful for dejavu users.

Equivalent to: [--TemplateExporter.exclude_input=False]

--embed-images

Embed the images as base64 dataurls in the output. This flag is only useful for the HTML/WebPDF/Slides exports.

Equivalent to: [--HTMLExporter.embed_images=True]

--sanitize-html

Whether the HTML in Markdown cells and cell outputs should be sanitized..

Equivalent to: [--HTMLExporter.sanitize_html=True]

--log-level=<Enum>

Set the log level by value or name.

Choices: any of [0, 10, 20, 30, 40, 50, 'DEBUG', 'INFO', 'WARN', 'ERROR', 'CRITICAL']

Default: 30

Equivalent to: [--Application.log_level]

--config=<Unicode>

Full path of a config file.

Default: ''

Equivalent to: [--JupyterApp.config_file]

--to=<Unicode>

The export format to be used, either one of the built-in formats

['asciidoc', 'custom', 'html', 'latex', 'markdown', 'notebook', 'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides', 'webpdf']
or a dotted object name that represents the import path for an
``Exporter`` class

Default: ''

Equivalent to: [--NbConvertApp.export_format]

--template=<Unicode>

Name of the template to use

Default: ''

Equivalent to: [--TemplateExporter.template_name]

--template-file=<Unicode>

Name of the template file to use

Default: None

Equivalent to: [--TemplateExporter.template_file]

--theme=<Unicode>

Template specific theme(e.g. the name of a JupyterLab CSS theme distributed as prebuilt extension for the lab template)

Default: 'light'

Equivalent to: [--HTMLExporter.theme]

--sanitize_html=<Bool>

Equivalent to: `[--NotebookExporter.nbformat_version]`

Examples

The simplest way to use nbconvert is

```
> jupyter nbconvert mynotebook.ipynb --to html
```

Options include ['asciidoc', 'custom', 'html', 'latex', 'markdown', 'notebook', 'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides', 'webpdf'].

```
> jupyter nbconvert --to latex mynotebook.ipynb
```

Both HTML and LaTeX support multiple output templates. LaTeX includes

'base', 'article' and 'report'. HTML includes 'basic', 'lab' and 'classic'. You can specify the flavor of the format used.

```
> jupyter nbconvert --to html --template lab mynotebook.ipynb
```

You can also pipe the output to stdout, rather than a file

```
> jupyter nbconvert mynotebook.ipynb --stdout
```

PDF is generated via latex

```
> jupyter nbconvert mynotebook.ipynb --to pdf
```

You can get (and serve) a Reveal.js-powered slideshow

```
> jupyter nbconvert myslides.ipynb --to slides --post serve
```

Multiple notebooks can be given at the command line in a couple of different ways:

```
> jupyter nbconvert notebook*.ipynb
> jupyter nbconvert notebook1.ipynb notebook2.ipynb
```

or you can specify the notebooks list in a config file, containing::

```
c.NbConvertApp.notebooks = ["my_notebook.ipynb"]
```

```
> jupyter nbconvert --config mycfg.py
```

To see all available configurables, use `--help-all`.

```
[124]: !ls -F
```

```
data/  sample_data/
```

```
[ ]:
```